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Valuations of epilepsy-specific health states: a comparison of patients with epilepsy and the general population

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ABSTRACT

Aims: Utility values that can be used in the economic evaluation of treatments for epilepsy can be elicited from the general population and the patient population, but it is unclear how the health state values differ. The aim of this study is to compare the preferences of the general population and a sample of people with epilepsy for health states described by the NEWQOL-6D QALY measure.
Methods: The Time Trade Off preference elicitation technique was used to value eight NEWQOL-6D health states. The general population sample was recruited and interviewed in their homes, and the sample with epilepsy was recruited and interviewed in an epilepsy service in North West England. Descriptive analysis and regression modeling were used to compare health state values across the populations.
Results: A sample of 70 people with epilepsy and a sample of 60 members of the general population were included. The populations differed across a range of background characteristics, but there were limited differences between the health state values. Patients provided significantly higher (better) values for the most severe health state described by the NEWQOL-6D ($p < 0.01$) and nonsignificant higher values for states with intermediate severity. The general population health state value was only higher for the best health state described by the NEWQOL-6D.
Conclusions: The similarities in the patient and general population values for NEWQOL-6D health states suggest that the use of the general population utility weights for the estimation of QALYs in the economic evaluation of epilepsy interventions is appropriate and largely representative of patient preferences.
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1. Introduction

Cost utility analysis (CUA) can be used to assess the cost effectiveness of treatments for epilepsy, with the quality adjusted life year (QALY) as the outcome measure. The QALY combines the health-related quality of life (HRQL) and time spent in a health state as a single figure to allow for comparisons across interventions and conditions. The quality aspect of the QALY is a figure anchored on a 0 (dead) to 1 (full health) scale described as the utility score. Utility values can be generated using many different measures, although the National Institute of Health and Care Excellence (NICE) recommends the use of EQ-5D [1] which has five dimensions of health (mobility, self-care, usual activities, pain/discomfort, anxiety/depression) scored according to one of three levels of severity (no problems, some or moderate problems, extreme problems). Utility scores for each of the 243 health states are derived from a tariff of values based on direct valuations of the general population using the Time Trade Off (TTO) method [2].

The generic nature of EQ-5D means that it is used across a wide range of patient groups, but the validity of the instrument has been questioned in epilepsy [3–6]. In response to these concerns, the authors developed an epilepsy-specific QALY measure from the NEWQOL instrument, the NEWQOL-6D [7]. The NEWQOL-6D (Fig. 1) assesses health across six dimensions (worry about attacks, depression, memory, concentration, control, stigma), each with four response levels, therefore describing 4096 (4^6) possible health states. To produce the utility values for use in the estimation of QALYs, a selection of health states was valued by a representative sample of the UK general population using the TTO preference elicitation technique. This produced a utility value set with a range from 0.341 (for the worst state) to 0.954 (for the best state).

Preferences for health states described by QALY measures that are used in the generation of utility values can be gained from both general population and patient samples. General population values are preferred by agencies such as NICE as it is argued that, where health care is publically funded, general public values should inform decision-making [8]. Evidence regarding differences in preferences for health

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Worry about attacks

You are not worried at all that you might have another epileptic attack

You are a little worried that you might have another epileptic attack

You are fairly worried that you might have another epileptic attack

You are very worried that you might have another epileptic attack

Depression

You never have problems with depression

You rarely have problems with depression

You sometimes have problems with depression

You always or often have problems with depression

Memory

You never have problems with your memory

You rarely have problems with your memory

You sometimes have problems with your memory

You always or often have problems with your memory

Concentration

You have no problem concentrating for more than a short period of time

You have mild problems concentrating for more than a short period of time

You have moderate problems concentrating for more than a short period of time

You have serious problems concentrating for more than a short period of time

Control

You feel that you have complete control over things that happen to you

You feel that you have some control over things that happen to you

You feel that you have little control over things that happen to you

You feel that you have no control over things that happen to you

Stigma

You do not feel that people treat you like an inferior person

You feel that some people maybe treat you like an inferior person

You feel that some people probably treat you like an inferior person

You feel that some people definitely treat you like an inferior person

Fig. 1. The NEWQOL-6D classification system.

states elicited from general population and patient samples is mixed, with evidence both for [9] and against [10] differences. This means that the choice of population may affect the values obtained and the subsequent economic evaluations carried out, but potential differences must be considered across different conditions. Values may differ across samples due to differences in interpretation and experience of living in and/or adapting to a particular health state or aspect of the state. Respondents are asked to imagine living in the health state presented, and the subsequent valuation is influenced by the individual's experience of their own or other people's health [11]. Values may differ based on whether the health state described relates to a physical or mental health condition, and this may be an important feature in epilepsy. However, no evidence regarding the valuation of epilepsy-specific health states across different populations is available.

The aim of this study was, therefore, to compare health state preferences from general population and patient samples using NEWQOL-6D health states. The aim was to understand how different populations perceive epilepsy-specific health states, and whether this impacts on the equivalence of health state values across the samples that could be used in the economic evaluation of epilepsy-specific interventions.

2. Methods

2.1. The Time Trade Off technique

Time Trade Off is used widely to value health states and was used to value NEWQOL-6D to promote comparability with EQ-5D, which was valued using a similar protocol [2]. This TTO protocol requires respondents to trade off a variable amount of years (x) in full health to avoid a fixed amount of ten years in a hypothetical health state described by the classification system. The trade-off follows an iterative process, until respondents are indifferent between the choices, where the value for the state is calculated as x/10. It is possible for respondents to indicate that they believe that the health state described is worse than being dead (i.e., where they would prefer 0 years in full health over 10 years in the health state), and, in these cases, the health state value is reached using a different method. Respondents choose between w years in the health state followed by x years in full health (where $w + x = 10$) and immediate death. A more detailed description of the valuation of NEWQOL-6D is provided in Mulhern et al. [7].

2.2. Health state selection

Health state descriptions were generated from the NEWQOL-6D classification system. The original NEWQOL-6D valuation study used an orthogonal array to select 50 health states for valuation which were allocated into 'blocks' of 8 states (which all included the worst state), with one block valued by each respondent. For this study, we used one of the blocks from the main valuation that included a wide range of health state severities and also the 'best' state (111111) described by the classification system.

2.3. Interview procedure

General population interviews were carried out in the respondent's home. Interviews with patients with epilepsy were conducted at a place convenient to participants; in most cases, this was in their own home. Alternatively, interviews were conducted at a meeting room within the Department of Public Health and Policy at the University of Liverpool and in a meeting room within the Neurology Clinic at the recruiting NHS Hospital Trust. At the start of the general population interview, respondents read the project information sheet to familiarize themselves with the aims and objectives of the study. Patients with epilepsy who met the study inclusion criteria (see Box 1) were given an information sheet in advance of the interview and contacted again within 48 h by one of the project researchers (DS). Potential participants were then able to discuss the project further and were given an opportunity to ask questions. If, following these discussions, patients with epilepsy wished to participate in the study, they were asked to sign a consent form, and arrangements to conduct the interview were formalized. At the commencement of the interview (both general population and patients with epilepsy), respondents self-completed the EQ-5D (with a one-day recall period) and the NEWQOL-6D (with a one-week recall period) classification systems to familiarize themselves with the health descriptions used in the valuation task (general population respondents did not complete the 'worry about epilepsy' dimension). Participants were then asked to rank the 8 health states to be valued plus the states 'full health' and 'dead' in order of preference from best to worst. This was done to familiarize participants with the health states. Following this, a practice TTO task using a hypothetical 'practice' state was completed. All respondents then valued the 8 NEWQOL-6D health states using TTO, with health states valued in a random order. At the end of the interview, respondents rated how difficult they found the rank and TTO tasks and answered questions about sociodemographic characteristics and health service use. Interviewers then completed a number of questions about the respondent understanding of the task and concentration levels.

Box 1. Study inclusion criteria: patients with epilepsy.

Study Inclusion criteria: Patients with Epilepsy

- Male or female
- Aged 18 years or over
- Have a confirmed clinical diagnosis of epilepsy
- Not diagnosed with a mental illness or learning disability
- Not participating in another research study
- Able to understand and converse in English
- Able to provide informed signed consent to participate

The general population interviews were carried out by Sheffield Hallam University, and the study was approved by the ScHARR Research Ethics Committee (Reference: CAO 21.01.10) at the University of Sheffield. The interviews of people with epilepsy were conducted by a member of the Liverpool University project team (DS), and the study was approved by the University of Liverpool Research Committee (Reference: UoL000566) and the National Research Ethics Service Committee North West — Liverpool Central (Reference: 10/H1001/3).

2.4. Study populations

The eight health states were valued by a subset of the sample from the original valuation study [7] and a sample of people with epilepsy. A sample size of at least 50 from each population was aimed to ensure sufficient power for a comparison of mean health state values across the different populations (assuming a power of 0.8, a significance level of 0.05, a standard deviation of 0.25, and an expected difference of 0.1).

2.4.1. General population

The general population sample was recruited by sampling households in urban and rural areas in North England using the AFD Names and Numbers version 3.1.25 database (AFD Software Limited, Ramsey, UK). Households that were sampled were first contacted by post, where the study was explained, and potential participants were asked to contact the research team if they were interested in taking part or if they did not want to be contacted again. One of four trained and experienced interviewers then visited households to request participation in the survey or to arrange an alternative time. Each sampled household was visited at different times of the week if initial contact could not be made. The block of health states used in this study was one of six used in the overall general population valuation to produce a full value set [7], and the administration of each block was rotated by interviewers to attempt to avoid demographic biases for any particular set of health states.

2.4.2. Patients with epilepsy

Patients with epilepsy were recruited from adult Mersey Region epilepsy clinic(s) held at the Walton Centre for Neurology and Neurosurgery. To ensure patient confidentiality, initial access to study participants was negotiated via members of the clinical health-care team who acted as 'gate keepers'. To ensure diversity among potential interviewees, participant selection was driven by those variables identified in the literature as having an impact on perceptions of the

quality of life of individuals with epilepsy. These variables included the following:

- Age
- Gender
- Age at onset (childhood/adulthood onset)
- Seizure type (tonic–clonic/partial).

Patients with epilepsy meeting the inclusion criteria (Box 1) were considered eligible.

2.5. Survey data collection and management

The Study Office in the School of Health and Related Research at the University of Sheffield was responsible for data checking, cleaning, coding, and entry. Anonymized data were stored securely on a password-protected database.

2.6. Analysis

Descriptive statistics and ANOVA were used to examine differences in respondent characteristics and mean health state values across the populations. Random effects generalized least squares regression analysis (which takes into account multiple observations per respondent) was used to examine the impact of both the severity of the health state and the population on elicited utility TTO values while controlling for sociodemographic characteristics. Three model specifications were estimated: (1) dummy variables for each health state and population, (2) Model 1 including dummy variables for background characteristics, and (3) dummy variables for each health state and dummy variables for the interaction between the population and the health states. Model 3 assessed whether there were interaction effects on the elicited TTO values and allowed us to understand the impact of population across the different health state severities. Performance of regression models was assessed using overall R-squared and Wald Chi-squared. The statistical programme STATA version 11 was used for all regression analyses, and SPSS version 19 was used for the descriptive statistical analysis.

3. Results

3.1. Respondents

Sixty members of the general population valued the eight NEWQOL-6D health states. The description of the recruitment and response rates of the overall general population sample is available in Mulhern et al. [9]. In total, one hundred and three (n = 103) patients with epilepsy agreed to speak with a member of the Liverpool research team (DS) to discuss the project in greater detail. Thirty-three (n = 33) declined subsequently to take part in the study. The final patient sample, therefore, included 70 people with epilepsy. Of this group, 53% reported tonic–clonic or tonic–clonic/other types of epilepsy, with the majority experiencing at least one seizure in the last year (73%) and a seizure in the last six months (68%).

The samples significantly differed in terms of gender (more males in the sample with epilepsy), marital status (more single people in the sample with epilepsy), and education level (lower education in the sample with epilepsy). The general population reported better health using the EQ-5D compared with the group with epilepsy (0.84 vs. 0.69); 15 (25%) reported experience of epilepsy in their family, and 21 (35%) reported experience of epilepsy as a carer (of those, 11 (18%) reported both). Nearly 25% of the sample with epilepsy reported as long-term sick (disabled), and none of the general population reported this. Significantly more of the sample with epilepsy self-reported that the TTO task was difficult to complete, and interviewers also reported that significantly more of the patient sample found the task difficult to understand.

3.2. Descriptive analysis

Mean and median health state values are reported in Table 1, with the health states ordered by the patient sample health state value. Analysis suggests modest differences between patient and general population values. Patients provided significantly higher values only for the most severe health state produced by the classification system (444444) ($F(1,117) = 6.89$, $p < 0.01$). The mean general population health state value was higher only for the best health state described by the classification system (111111), but the difference is not significant ($F(1,125) = 0.20$, $p = 0.69$). Patients provide higher values for the six intermediate health states, but the differences are less than 0.05 and are not significant (p range = 0.42 to 0.98). There is some evidence that the samples ranked the severity of the health states differently, as the order of the TTO values provided differs for the four intermediate states (between 3 and 6 in the ranking). However, the

Table 1
Demographic characteristics of the samples.

Characteristics	Patient	General population	Significance
n	70	60	
Age	43.47 (14.61)	46.03 (16.22)	
Gender			<0.001
Male	31 (44.3)	20 (33.3)	
Female	39 (55.7)	40 (66.7)	
Marital status			<0.001
Single	23 (32.9)	10 (16.7)	
Married/partner	35 (50.0)	41 (68.3)	
Separated	2 (2.9)	1 (1.7)	
Divorced	8 (11.4)	7 (11.7)	
Widowed	2 (2.9)	1 (1.7)	
Employment status			<0.001
In employment	29 (41.4)	35 (58.3)	
Retired	10 (14.3)	11 (18.3)	
Housework	2 (2.9)	5 (8.3)	
Student	3 (4.3)	4 (6.7)	
Seeking work	3 (4.3)	2 (3.3)	
Unemployed	1 (1.4)	1 (1.7)	
Long-term sick	17 (24.3)	0	
Other	4 (5.7)	2 (3.3)	
Education postminimum			0.128
Yes	46 (65.7)	42 (70.0)	
No	24 (34.3)	18 (30.0)	
Degree level			<0.001
Yes	30 (42.9)	18 (30.0)	
No	40 (57.1)	42 (70.0)	
EQ-5D			<0.001
Index score (mean (sd))	0.689 (0.30)	0.843 (0.22)	
Range	−0.181 to 1	−0.181 to 1	
Experience of epilepsy			
Yourself		1 (1.6)	
Your family		15 (25.0)	
As a carer		21 (35.0)	
Task difficulty			<0.001
Difficult	34 (48.6)	20 (33.3)	
Neither difficult nor easy	12 (17.1)	15 (25.0)	
Easy	24 (34.3)	25 (41.7)	
Some problems/doubtful respondent understood task	26 (37.1)	15 (25.0)	<0.001
Patient sample			
Type of epilepsy			
Tonic–clonic	9 (12.9)		
Tonic–clonic and other types	28 (40.0)		
Other types only	33 (47.1)		
Seizure frequency (last year)			
None	19 (27.1)		
1–9	23 (32.8)		
10+	28 (40.0)		
Time since last seizure			
Last week	24 (34.8)		
Last month	8 (11.6)		
1–6 months ago	15 (21.7)		
>6 months	19 (27.5)		
Do not know	3 (4.3)		

Table 2
Descriptive statistics of NEWQOL-6D health state values by population.

State	Patient					General population				
	n	Mean (sd)	Median	Range	Rank	n	Mean (sd)	Median	Range	Rank
111111	67	0.929 (0.15)	1	0.3 to 1	1	60	0.942 (0.18)	1	0 to 1	1
113132	67	0.847 (0.19)	0.925	0 to 1	2	60	0.827 (0.22)	0.925	0 to 1	2
223333	68	0.822 (0.19)	1	0.3 to 1	3	60	0.793 (0.21)	0.85	0 to 1	4
224312	66	0.814 (0.20)	0.875	0.275 to 1	4	60	0.812 (0.28)	0.925	−0.475 to 1	3
332222	67	0.794 (0.29)	0.90	−0.675 to 1	5	60	0.766 (0.26)	0.825	0 to 1	6
143113	67	0.779 (0.22)	0.825	0 to 1	6	62	0.771 (0.29)	0.875	−0.675 to 1	5
144422	54	0.650 (0.34)	0.70	−0.775 to 1	7	59	0.640 (0.22)	0.70	−1 to 1	7
444444	65	0.519 (0.35)	0.50	−0.2 to 1	8	60	0.360 (0.32)	0.35	−0.5 to 1	8

rankings for the two most mild and two most severe states do not differ by population, and the severity of the four states where the ranking is not consistent is relatively similar in terms of the health state values provided, so some differences may be expected. Within the general population sample, there were no differences in the mean health state values provided across respondent groups with ($m = 0.704$) or without ($m = 0.720$) past experience of epilepsy in themselves, in their family, or in caring for others ($F(1,59) = 0.29$, $p = 0.61$). (See Table 2.)

3.3. Regression analysis

Model 1 shows that each health state impacts significantly on the TTO value provided in comparison with the best health state described by the classification system (111111). However, the population which the respondent belongs to is not statistically significant. Model 2 shows that the only significant background variable is past experience of epilepsy in the respondent's family (for the general population sample only). Model 3 shows that only the interaction between the worst health state (444444) and the population providing the value is significant, as is suggested by the mean differences in the values outlined in Section 3.2 (See Table 3.).

Table 3
Regression results.

	1	2	3
States			
111111	Ref	Ref	Ref
113132	−0.094***	−0.106**	−0.114***
223333	−0.127***	−0.156***	−0.149***
224312	−0.118***	−0.139***	−0.129***
332222	−0.151***	−0.181***	−0.176***
143113	−0.161***	−0.175***	−0.177***
144422	−0.287***	−0.304***	−0.301***
444444	−0.492***	−0.601***	−0.581***
Patient	0.034		
111111 × patient			−0.013
113132 × patient			0.026
143113 × patient			0.016
144422 × patient			0.014
223333 × patient			0.031
224312 × patient			0.008
332222 × patient			0.034
444444 × patient			0.158***
Female		−0.039	
Married		−0.058	
Employed		0.041	
Min education level		0.000	
Age			
18–40		Ref	
41–65		−0.072	
65+		−0.187	
Experience of epilepsy (GP sample)			
Your family		0.168**	
As a carer		0.007	
Constant	0.917***	1.483***	0.941***
N obs	1010	1010	1010
R^2 overall	0.23	0.37	0.24

Asterisks are for significance at the 0.001 level

4. Discussion

We have reported the results of the first comparison of valuations by the general population and patients with epilepsy of epilepsy-specific health states and found limited differences between the values provided. This means that utility values derived from the general population for use in the estimation of QALYs may also generally represent the views of people with epilepsy. However, there may be differences at the more severe end of the scale, where general population respondents overestimate the perceived HRQL impacts of the state in comparison with people with epilepsy who may or may not have experienced a similar health state to those described by the NEWQOL-6D. This may mean that if a full valuation was carried out with people with epilepsy, the value of the worst states could be higher. A possible explanation for this could be a higher level of perceived adaptation to the health state over time in the TTO task among people with epilepsy [8]. Also, the use of TTO as the preference elicitation task may impact on this, as people with experience of the health descriptions may be unwilling to sacrifice years of life to avoid health states similar to their own.

The results of past work into differences between patient and general population values are inconclusive, with a meta-analysis finding limited differences [10], but other studies investigating both physical [12] and mental health conditions including dementia [9] and depression [13] finding that patients valued the health states differently. One reason for the similarities in values reported here may be because the general population is more aware of the HRQL impacts of epilepsy as measured by the NEWQOL-6D than other conditions. Furthermore, a significant proporation of the general population sample had experience of epilepsy either in their family or in caring for others which may impact on the values given to health states.

The similarity of the values across the populations suggests that a patient value set would not be significantly different from the existing general population value set, and both could be used to allocate resources. Our findings do not challenge the preference of decision-making bodies such as NICE for the use of general population value sets to provide the quality component of the QALY. This may not be the case with more generic instruments such as the EQ-5D, which is the measure preferred by NICE [1], and further research could investigate how patients with epilepsy value nonepilepsy-specific health states described by the EQ-5D. This may inform the debate about whose values should be used in the allocation of resources across conditions.

There are a number of differences in the demographics of the two samples, and the perceived difficulty of the TTO task to respondents. For example, although the two samples are of similar age, less of the patient sample are in employment (with more classified as long-term sick (disabled)), and less are married or with a long-term partner. Significantly more of the patient groups are educated to at least degree level, but more found the TTO task difficult, which may reflect the focus of

the task which asks specifically about health states related to the condition that they have. However, that regression analysis indicates that these differences do not lead to significant differences in preferences for health states. A substantial proportion of the general population sample also reported past experience of epilepsy. This may be linked to the recruitment process which informed participants that the study was about epilepsy although this does not seem to impact on the health state values provided.

This study has a number of limitations which restrict the conclusions that can be drawn from the data. The small number of states included in the study means that we cannot model the results to produce a utility value for every health state described by the NEWQOL-6D using the preferences of people with epilepsy and can only use the mean values from a small number of states to conclude that the overall scales would be similar. However, the states included in this study cover the full severity range, and each severity level of each dimension is represented at least once. Further information about the sample with epilepsy (for example, more detailed data about type of epilepsy or seizure control) may have allowed for more in-depth analysis about whether the health state values provided differ across these groups. However these data were not available to the research team.

In conclusion, the similarities in the patient and general population values for NEWQOL-6D health states mean that value sets from both populations may be comparable. The NEWQOL-6D general population value set can be used to complement the EQ-5D in the economic evaluation of epilepsy interventions.

Conflict of interest

The authors do not have any conflicts of interest.

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